Module 3: Organic Molecules - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Explain why carbon is central to biological molecule diversity.
2. Distinguish monomers, polymers, dehydration synthesis, and hydrolysis.
3. Compare the functions of carbohydrates and lipids.
4. Connect protein shape to protein function and denaturation.
5. Describe how nucleic acids encode biological information.

Topics

- Carbon skeletons
- Building and breaking polymers
- Carbohydrates and lipids
- Proteins and shape
- Nucleic acids and information

Key Terms to Know

- **Organic molecule** - Carbon-based molecule associated with living systems.
- **Monomer** - Small building block that can join into a polymer.
- **Polymer** - Large molecule built from repeating subunits.
- **Carbohydrate** - Sugar or sugar polymer used for energy or structure.
- **Lipid** - Hydrophobic molecule including fats, phospholipids, and steroids.
- **Protein** - Polymer of amino acids that performs cellular work.
- **Nucleic acid** - DNA or RNA polymer made of nucleotides.
- **Denaturation** - Loss of protein shape that often destroys function.

Core Contents

1. Carbon forms diverse skeletons that make biological molecules possible.
2. Dehydration synthesis builds polymers, and hydrolysis breaks them down.
3. Carbohydrates and lipids store energy and provide structure in different ways.
4. Protein function depends on amino acid sequence and three-dimensional shape.
5. DNA and RNA store and move information using nucleotide sequences.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the module's companion lab (see the course schedule pairing) as the evidence check.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-04_liquid-chemistry.md`

Generated Visuals

- Module 03: Organic Molecules Evidence Map
- Module 03: Organic Molecules Reasoning Sequence
- Module 03: Organic Molecules Retrieval and Lab Check